

GROUP WORK PROJECT # __2_
DATA
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Q1. Data Types Used for Stock Market Movement Prediction

Primary Data Sources and Collection

The researchers used historical price data for three Exchange-Traded Funds (ETFs) obtained from Yahoo Finance:

- iShares MSCI Chile ETF (ECH) - an emerging market ETF
- iShares MSCI Brazil ETF (EWZ) - an emerging market ETF
- iShares Core S&P 500 ETF (IVV) - a developed market ETF for comparison

The data covered the period from December 12, 2009, to January 1, 2020. This timeframe was deliberately selected to exclude market distortions caused by the COVID-19 pandemic, focusing on more typical market behavior patterns.

Raw Data Components

For each ETF, the following daily price data points were collected:

- Opening price (Open)
- Highest price (High)
- Lowest price (Low)
- Closing price (Close)
- Trading volume (Volume)
- Adjusted closing price (Adjusted Close) - adjusted for splits, dividends, and capital gain distributions

Technical Indicator Derivation

From these six fundamental data points, the researchers expanded their feature set dramatically:

- They used the Pandas Technical Analysis Library (Pandas TA) to calculate 210 additional technical indicators
- Combined with the 6 original features, this created a total of 216 daily features for analysis
- These technical indicators fall into various categories including: Candles, Cycles, Momentum, Overlap, Performance, Statistics, Trend, Utility, Volatility, and Volume

Target Variable Construction

The researchers created a binary classification target variable (Γ) defined as:

- $\Gamma(t) = 1$ if $\text{Open}(t) - \text{Open}(t-1) > 0$ (price increased)

- $\Gamma(t) = -1$ otherwise (price decreased or remained the same)

This transformed the prediction task into a binary classification problem predicting whether the opening price would rise or not on the following day.

Importance of Technical Indicators in Forecasting Stock Price Trends

Addressing Market Complexity

The paper emphasizes that stock market prediction is challenging due to the "energetic, nonlinear, nonparametric, and chaotic properties of stock information." Technical indicators help address this complexity by:

- Capturing patterns that might not be evident in raw price data
- Providing quantitative measurements of market psychology and momentum
- Revealing different aspects of market behavior across various timeframes

Q2. Security Understanding

The iShares MSCI Chile ETF (ECH) is an exchange-traded fund that tracks the performance of the Chilean equity market by replicating an index composed of Chilean stocks. As an emerging market ETF, it provides investors with exposure to the Chilean economy without requiring direct investment in individual Chilean securities. ECH is managed by BlackRock, one of the world's largest asset management companies.

Asset Type and Composition

ECH is composed primarily of large and mid-capitalization Chilean equities. According to the research paper, the fund has significant sector concentration with the following distribution:

- Financials: 21.53%
- Materials: 21.28%
- Utilities: 18.73%
- Consumer Staples: 13.92%
- Energy: 8.34%

Price History and Performance

The price history of ECH from 2009 to 2020 (the period analyzed in the study) shows considerable volatility typical of emerging markets:

- Minimum price: \$29.30
- Maximum price: \$80.25
- Median price: \$46.48
- Mean price: \$50.10

Trading Statistics

The study reports the following statistical measures for ECH's opening price:

- 1st Quartile: \$40.35
- 3rd Quartile: \$59.84
- Price Range: \$50.95 (from min to max)

Why Classification Rather Than Regression?

1. **Directional focus:** The paper emphasizes that for many investors, the direction of price movement (up or down) is more important than the exact magnitude of the change. Classification provides this binary signal directly.
2. **Reduced complexity:** Stock prices are highly volatile and affected by numerous factors, making precise numerical predictions extremely challenging. By simplifying to directional prediction, the model becomes more robust.
3. **Performance evaluation:** Classification accuracy is easier to evaluate than regression errors when dealing with highly volatile time series data.
4. **Practical application:** For trading strategies, especially in emerging markets with higher volatility, knowing the direction often provides sufficient information for decision-making.

Alternative Classification Definitions

The authors defined their classification variable Γ based on whether the opening price increased from the previous day. Two alternative approaches could be:

1. **Volatility-adjusted price movement:** $\Gamma(t) = 1$ if $(Close(t) - Open(t))/\sigma > \text{threshold}$ $\Gamma(t) = -1$ otherwise

Where σ represents the standard deviation of price changes over a rolling window. This would classify days based on whether the price movement exceeds typical volatility, identifying more significant moves and filtering out noise.

2. **Volume-weighted price direction:**

$$\Gamma(t) = \begin{cases} 1, & \text{if } (Close(t) - Open(t)) > 0 \text{ and } Volume(t) > \text{Threshold} \\ -1, & \text{if } (Close(t) - Open(t)) < 0 \text{ and } Volume(t) > \text{Threshold} \\ 0, & \text{otherwise} \end{cases}$$

This classification would give more weight to price movements accompanied by higher trading volumes, potentially identifying more significant market moves with stronger conviction, as higher volume often indicates stronger market sentiment behind a price movement.

Q3. Methodology Understanding

The original "Materials and Methods" section can be separated into two distinct sections. Here's how Section 2 (Data) should be organized:

2. Data

2.1 Stocks Analyzed

- Selection of ETFs (ECH, EWZ, IVV)
- Time frame rationale (December 12, 2009, to January 1, 2020)
- Market exposure characteristics
- Pre-pandemic data importance for feature selection

2.2 Data Sources

- Yahoo Finance data components (Open, High, Low, Close, Volume, Adjusted Close)
- Descriptive statistics of the opening price data

2.3 Data Processing

- Definition of the target variable Γ (class label for prediction)
- Data normalization using min-max approach
- Data cleaning (handling missing values)
- Class assignment methodology

2.4 Technical Indicators

- Implementation of Pandas Technical Analysis Library
- Extension of original 6 attributes to 216 daily features
- Types of indicators calculated (momentum, trend, volatility, etc.)

New Section 3: Methodology

The methodology section should focus on the analytical approaches and techniques used:

3. Methodology

3.1 Cross-Industry Standard Process for Data Mining (CRISP-DM)

- Six-stage methodological framework
- Implementation in the ETF prediction context

3.2 Feature Selection Methods

- Low Variance approach
- Chi-Squared technique
- LASSO (Least Absolute Shrinkage and Selection Operator)
- Tree-based Feature Selection
- Principal Feature Analysis
- Mean Absolute Difference (MAD)
- Dispersion Ratio (DR)

3.3 Neural Network Model

- Multilayer Perceptron (MLP) architecture
- Configuration parameters for prediction
- Training process details

3.4 Cross-validation

- K-fold cross-validation process (K=10)
- Algorithm implementation for validation

3.5 Experimental Design

- Comprehensive experimental methodology (Algorithm 2)
- Feature selection optimization process
- Model evaluation approach

Dividing Descriptive Statistics from Models

The distinction between descriptive statistics and models could be made as follows:

Descriptive Statistics:

- Pearson's Correlation (measures relationships between variables)
- Mean Absolute Difference (describes data dispersion)
- Dispersion Ratio (measures feature relevance)
- Low Variance approach (identifies constant features)

Models:

- LASSO (a regularization technique for feature selection and model fitting)
- Tree-based Feature Selection (machine learning approach for feature importance)
- Principal Feature Analysis (dimensionality reduction technique)
- Chi-Squared (statistical test for independence)
- Neural Network Model (predictive model)

Technical Indicators Optimization Process

The optimization process for technical indicators in the paper follows these steps:

1. **Feature Calculation and Preprocessing:** The authors calculate 216 technical indicators for each ETF, normalize the data, and clean it by removing days with missing values.
2. **Statistical Feature Selection:** Multiple statistical measures are applied to identify salient features, including correlation analysis, MAD, DR, and other techniques.
3. **Feature Subset Creation:** The algorithm identifies features that appear in at least a specified number of subsets defined by statistical measures, creating different Selected(n) subsets.
4. **Iterative Evaluation:** The MLP model is trained and evaluated on different subsets of features using K-fold cross-validation (K=10).
5. **Performance Measurement:** Average accuracy across cross-validation folds is calculated for each subset of features.

The importance of this optimization process is highlighted in the results, where the Selected(5) subset achieves better prediction accuracy (77.82-80.27%) while using only 4.16-5.09% of the original features. This significantly reduces computational resources and training time (by 84.68% on average) while improving accuracy by approximately 13.63%.

The optimization is particularly valuable because:

- It identifies the most relevant features for each specific ETF
- It removes redundant information that could lead to overfitting
- It creates more efficient models with fewer parameters
- It improves generalization by focusing on the most predictive indicators
- It significantly reduces computational costs while maintaining or improving performance

Q4. Feature Understanding

Sagaceta Mejia et al. (2024) describes a feature as a technical indication that is computed from stock price and volume data and used as input to the machine learning model. These technical indicators, which are based on historical trade data, include widely used metrics including Moving Averages (MA), Relative Strength Index (RSI), Momentum (MOM), On-Balance Volume (OBV), Rate of Change (ROC), and others (Sagaceta Mejia et al., 2024).

In essence, a feature is a processed data point extracted from raw financial information to enhance model learning.

a) How do you distinguish a feature from a method and from a model?

- A feature is an input variable or other numerical representation that has been engineered or extracted from raw data, and is used in predictive modeling and statistical analysis.
- A method is a procedure or approach, such as Principal Component Analysis (PCA) or LASSO regression that is useful in selection or modification.
- A model is a trained algorithm that makes predictions based on features. A commonly used machine learning model in stock market prediction is neural networks, which have proven helpful in discovering nonlinear and nonadditive data relations, attaining superior results (De Haan et al., 2016).

Aspect	Feature	Method	Model
Example	RSI(14-day), 50-Day Moving Average	Feature Selection via LASSO	Neural Network Classifier

(Source: Sagaceta Mejia et al., 2022; James et al., 2013)

b) What are the categories of features that you have learned?

Features can be grouped as follows:

- **Price-based indicators:** e.g., Moving Averages (SMA, EMA).
- **Momentum indicators:** e.g., RSI, MACD.
- **Volume-based indicators:** e.g., On-Balance Volume (OBV).
- **Volatility indicators:** e.g., Average True Range (ATR).
- **Trend-following indicators:** e.g., Directional Movement Index (DMI).

These categories are critical as they capture different market behaviors: trends, reversals, or trading volumes (Achelis, 2001).

c) Optimization of Technical Indicators

The study maximizes the predictive ability of technical indicators by modifying their parameter settings, such as time periods. After LASSO (Least Absolute Shrinkage and Selection Operator) regression is used to penalize irrelevant features, the neural network's hyper parameters are tuned (Sagaceta Mejia et al., 2022).

Optimizing indicators reduces over fitting, strengthens signal extraction, and boosts out-of-sample prediction accuracy, which is essential in volatile emerging markets.

Q5. Optimization Understanding

a. Cross Validation in Words

Cross-validation is a resampling technique used to evaluate predictive models by partitioning the original sample into training and testing sets multiple times. This ensures that the model's evaluation is less dependent on a particular data split and gives a more generalized measure of performance (James et al., 2013).

Cross-validation methods can be used to estimate the accuracy of such models on new patients originating from the same source, by repeated random splitting of the data (Leinonen et al., 2024).

b. k-fold cross validation in words?

The process of the k-Fold cross-validation is explained as follows:

- Divide the data into k folds.
- Out of the k folds, k-1 sets are used for training whereas the remaining set is used for testing.
- The model is then trained and tested k times, each time a new set is used as testing whereas the remaining sets are used for training.
- Lastly, the result of the k-Fold cross-validation is the average of the results obtained on each set.

A reliable assessment of model accuracy is then provided by averaging the performance metric over all k trials (Hastie, Tibshirani, & Friedman, 2009).

c. Jaccard Distance>

The Jaccard Distance is a frequently employed metric to evaluate the similarity or dissimilarity between sets.

The Jaccard distance $J \delta$ is known to fulfill all properties of a metric, notably, the triangle inequality—a fact that has been observed many times, e.g., via metric transforms (Kosub, 2019).

This measure provides a simple and easy way to quantify the difference across sets, making it a useful tool in a variety of disciplines where determining similarity or dissimilarity is crucial. The Jaccard Distance is defined as follows:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

Its range always lies between 0 and 1, with 0 signifying complete similarity, and 1 representing complete dissimilarity.

The Jaccard Distance is used in this paper to evaluate the similarity between the predicted and true class labels, especially useful in multi-class classification tasks (Mejia et al., 2022).

d. Comparison of Jaccard Distance with two other metrics

Metric	Definition	Application
Jaccard Distance	Dissimilarity between two sets	Best for classification outcomes
Euclidean Distance	Straight-line distance between points	Best for continuous numeric prediction (regression)
Manhattan Distance	Sum of absolute differences between points	Better in high-dimensional data, robust to outliers

(Source: Friedman, Hastie, and Tibshirani, 2009)

Jaccard distance is more suited for category or set-based comparisons, like forecasting upward or downward stock movement, whereas Euclidean and Manhattan distances work well for continuous-valued data.

e. Definition of Optimal Solution by authors

The optimal solution is defined as the combination of features and model hyper parameters that **maximize the prediction accuracy** (or minimize the error) on unseen validation data, as measured through metrics like the Jaccard index and accuracy percentage (Mejia et al., 2022).

The optimal solution is the one that satisfies all the constraints and yields the best value for the objective function, according to the investor's goals—such as maximizing return or minimizing risk (Cornuéjols et al., 2018).

In financial contexts, optimality must account not only for current conditions but for all possible future scenarios, weighted by their likelihood (Ziemba et al., 2014).

An optimal solution in finance is often approximate, computed through numerical

methods that seek to minimize error and maximize convergence reliability (Gilli et al., 2011)

Step 1: Financial Problem

The Financial Problem the Authors Aim to Solve

Based on Section 3 of the paper, the authors aim to address the challenge of predicting ETF (Exchange-Traded Fund) trend directions in emerging markets with greater accuracy and computational efficiency. Specifically, they focus on:

1. Identifying the most salient technical indicators that effectively predict ETF price movements in emerging markets
2. Reducing computational costs while maintaining or improving prediction accuracy
3. Providing investors with a reliable tool for market timing and risk management in emerging markets
4. Enabling better decision-making for entering or exiting positions in emerging market ETFs

As stated in Section 3.1 (Practical Implications for Investors in Emerging Markets), their model helps investors with "market timing and risk management within emerging markets" and allows them to "ascertain the optimal timing for entering or exiting positions in ETFs, effectively maximizing returns and minimizing losses."

Differences Between Predicting Emerging vs. Developed Markets

The paper reveals several key differences between predicting stock movements in emerging versus developed markets:

1. **Different Technical Indicators:** The analysis shows that emerging markets depend more on cyclical behaviors of prices, while developed markets do not exhibit this same dependence. This is evident in the feature selection results where "Cycles" category features appear in both emerging market ETFs (ECH and EWZ) but not in the developed market ETF (IVV).
2. **Quantitative vs. Qualitative Features:** The authors note that "emerging markets' predictions use quantitative features, while developed markets rely more on qualitative features." For example, TTM Trend (a qualitative indicator) is only selected for the developed market ETF (IVV).
3. **Volume Indicators:** Different volume indicators are relevant for each market type. For emerging markets, Archer's On Balance Volume (AOBV) is selected, while Price Volume Rank (PVR) is selected for developed markets. These indicators measure volume-price relationships differently.
4. **Market Exposure Differences:** The paper suggests that market exposure (sector distribution) influences which indicators are most predictive. Table 1 shows significantly different sector distributions between emerging market ETFs and the developed market ETF.
5. **Volatility Patterns:** Emerging markets exhibit higher volatility and different risk characteristics, requiring specialized feature selection for effective prediction.

This distinction is significant for the model's design because it demonstrates that a one-size-fits-all approach to technical analysis is suboptimal. The selection of features must be tailored to the specific market type, with different indicators having varying predictive power in emerging versus developed markets.

Step 2: Application

Main Takeaways of the Results

1. **Feature Reduction Success:** The most impressive result is that using only 5% of the original features (the Selected(5) subset) achieves better prediction accuracy than using all 216 features. This demonstrates the importance of feature selection in financial prediction models.
2. **Accuracy Improvement:** The Selected(5) feature subset achieved prediction accuracy between 77.82% and 80.27%, which was higher than using the complete set of features.
3. **Computational Efficiency:** When using early stopping techniques with the reduced feature set, the training time was reduced by 84.68% on average while improving accuracy by 13.63%.
4. **Market Similarities:** The Jaccard Distance analysis showed that ETFs from emerging markets (ECH and EWZ) had more similar feature sets ($J=0.33$) compared to comparisons with the developed market ETF ($J=0.64$ and $J=0.53$), suggesting common characteristics across emerging markets.

Specific Useful Features from the Study

The most useful features identified across ETFs were:

1. **BBP_5_2.0** (Bollinger Band Percent): Appeared in all three ETFs, quantifying price relative to Bollinger Bands
2. **BOP** (Balance of Power): Selected for all ETFs, measures systematic buying/selling pressure
3. **DEC_1** (Decreasing): Boolean indicator of price decrease compared to previous period
4. **INC_1** (Increasing): Boolean indicator of price increase compared to previous period
5. **J_9_3** (KDJ Indicator): Appears in all ETFs, analyzes and predicts changes in stock trends

For emerging markets specifically:

1. **AOBV_LR_2** (Archer's On Balance Volume): Volume indicator for both emerging market ETFs
2. **CTI_12** (Correlation Trend Indicator): Measures correlation of price with trend line
3. **EBSW_40_10** (Even Better SineWave): Measures market cycles using a low-pass filter
4. **K_9_3**: Component of the KDJ indicator

For the developed market (IVV):

1. **PVR** (Price Volume Rank): Compares direction of price change to volume change
2. **TTM_TRND_6** (TTM Trend): Identifies price bars above/below average
3. **STOCHRSI_14_14_3_3** (Stochastic RSI): Applies stochastic oscillator to RSI values.

Step 3: Replication

Abstract

Integrating big data analytics and machine learning algorithms has become increasingly important in the fast changing landscape of stock market investment (Sagaceta-Mejia, 2024). This replication study tries to investigate the predictive power of lagged daily returns on next-day price indication using data from the iShares Core S&P 500 ETF (IVV). Drawing on the methodology presented by Sagaceta Mejia et al. (2024), we construct a binary classification target representing next-day return movement and apply a 5-fold cross-validation framework to assess the relationship between the lagged return and the target. We used correlation coefficients to quantify this relationship across 5-folds. We get an average correlation of -0.0277. These results suggest that lagged return alone might not be sufficient for accurate market forecasting and shows the importance of technical indicator importance and feature selection. Identifying and selecting the most salient features (technical indicators) is critical in predicting the

trend direction of exchange-traded funds (ETFs) in emerging markets, leveraging financial and economic indicators (Sagaceta-Mejia et al., 2024).

Keywords: ETF, technical indicators, feature selection, lagged return

1. Introduction

An exchange-traded fund (ETF) is a relatively recent financial innovation that provides an alternative method for indirectly investing in international equities (Sagaceta-Mejia et al., 2024). ETFs have a long history and they started trading in US in 1993. They were developed as an alternative to mutual funds. ETFs have often promoted as better than mutual funds due to their lower fees, trading flexibility and diversification. This fund allows investors to implement different investment strategies incorporating diverse geographic and economic activities rather than investing in local handmade portfolios (Deville, 2008; Antoniewicz and Heinrichs, 2014).

The accurate prediction of the stock market remains a fundamental yet highly challenging objective in financial research due to the volatility, noise, and stochasticity in financial markets. A central theoretical debate in financial economics revolves around the effectiveness of technical indicators in predicting prices (Akash et al., 2025). Among these indicators, historical returns are frequently employed as stand-ins for momentum, a notion derived from assets' propensity to keep advancing in the same direction. However, the effectiveness of such indicators has continued to be a strong subject of debate especially in the context of efficient modern markets.

Sagaceta Mejia et al. (2024) present a methodology that encompasses an array of statistical techniques strategically employed to identify critical technical indicators with significant implications for time series problems. They develop a more accurate trend prediction mechanism by performing systematic evaluations of statistical and machine learning methods across multiple sets of features; feature selection via LASSO and the tuning of technical indicator parameters to improve predictive power.

The purpose of this replication study is to identify and assess, in a more straightforward setting, the impact of a single, **unoptimized indicator—lagged return**. To find out if such a basic predictor has any meaningful signal, we apply the method to iShares Core S&P 500 ETF (IVV) - a developed market ETF that represents the US equities market. The IVV is a highly liquid and developed market asset; and its data is readily available.

2. Methodology

To increase the reliability in our findings we used a range of data from January 1, 2010 to January 1, 2020 as proposed by Sagaceta Mejia et al., (2024). This time frame is

ideal; it allows us to exclude economic changes caused by atypical phenomena during global pandemics (Almehmadi, 2021; Verma et al., 2022) that would otherwise greatly affect the results of our study.

We retrieved historical daily price data for IVV from Yahoo Finance. The primary feature used was the one-period lagged daily return, calculated from adjusted closing prices. We constructed a binary target variable such that a value of 1 indicated a positive next-day return and 0 otherwise. No other technical indicators or external data were included. We did not perform any feature optimization as well.

We used a 5-fold cross validation to analyze the relationship between the lagged return and the binary target.

The process of the 5-Fold cross-validation is explained as follows:

- Divide the data into 5 folds.
- Out of the 5 folds, 5-1(=4) sets are used for training whereas the remaining set is used for testing.
- The model is then trained and tested 5 times, each time a new set is used as testing whereas the remaining sets are used for training.
- Lastly, the result of the 5-Fold cross-validation is the average of the results obtained on each set.

For each fold, we computed the point-biserial correlation coefficient, a metric appropriate for assessing the relationship between a continuous variable and a binary classification target. This procedure was implemented in Jupyter Notebook using the yfinance, pandas, scikit-learn, and scipy libraries.

3. Results

The results obtained with the methodology previously described are condensed in Table 1, graph 1 and histogram 1.

Table 1 summarizes the point-biserial correlations observed in each of the five validation folds:

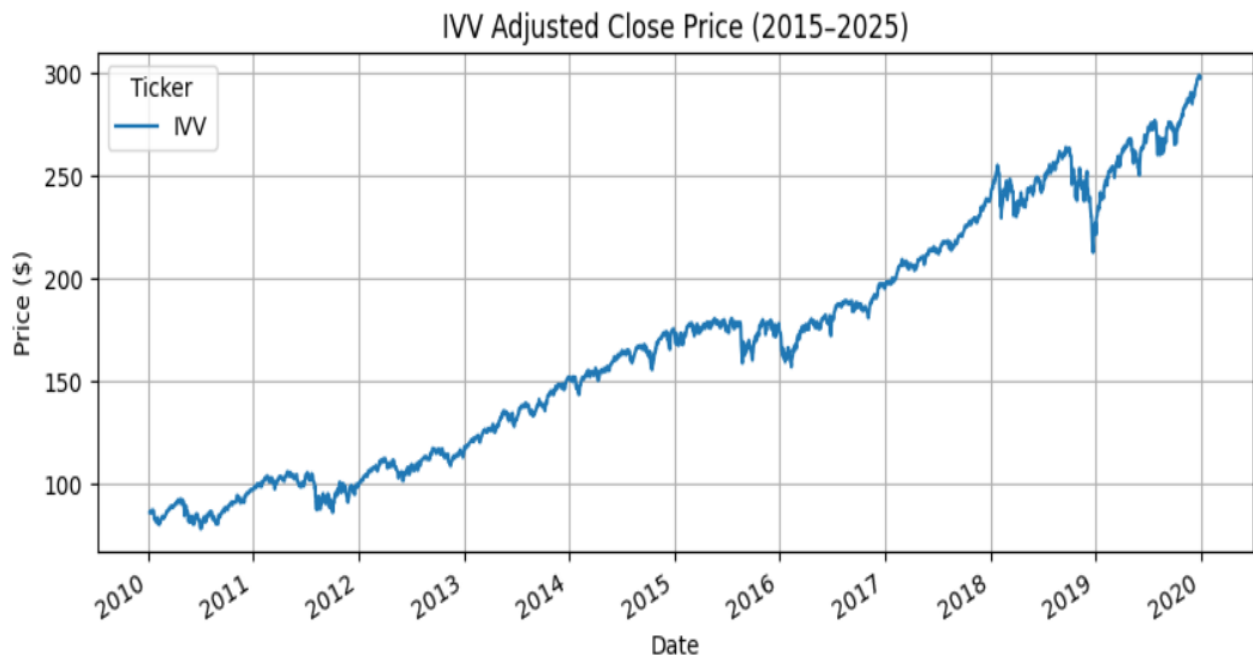
Table 1: Cross-Validation Results Table

Fold	Correlation
1	-0.008314
2	-0.023275
3	0.007699
4	-0.082747
5	-0.032265
Average	-0.032265

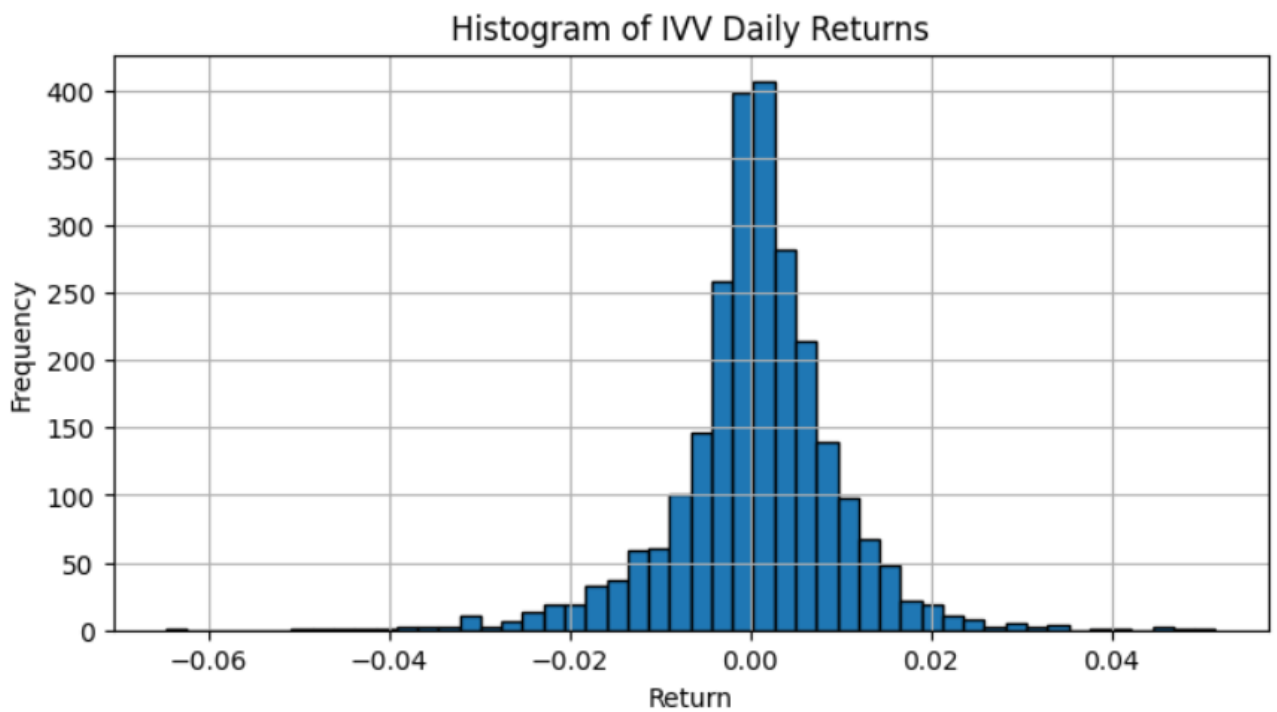
The average correlation of -0.032265 demonstrates a weak and inconsistent relationship between lagged returns and next-day price direction. The correlation fluctuated across folds, as shown in Table 1; only fold 3 had a slightly positive correlation. On average, the relationship was slightly negative, but of such a small magnitude that it is unlikely to have meaningful predictive power.

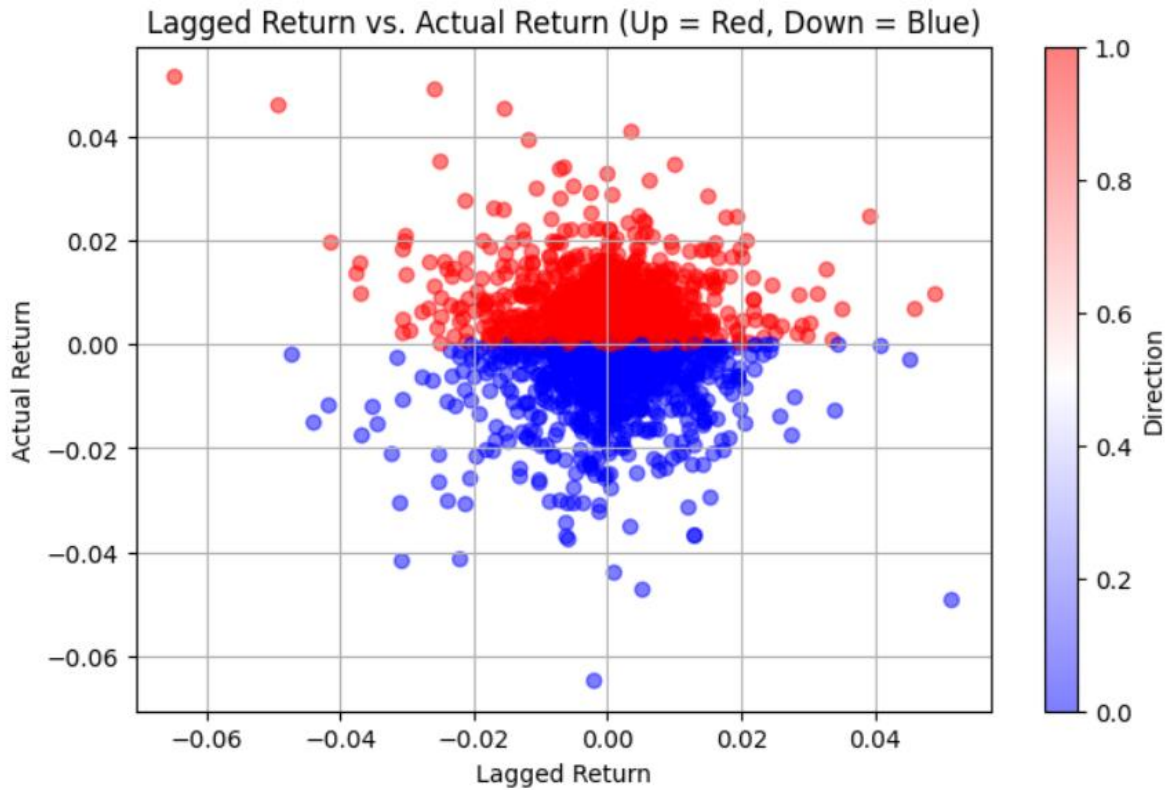
The visuals support the statistical interpretation much more. The time series plot of IVV's adjusted close prices from 2010 to 2020 shows short-term volatility across a typically increasing trend that is consistent with the performance of the U.S. equities market over the long run. The histogram of daily returns supports the empirical characteristics of financial return series, which include larger tails than the Gaussian benchmark and a near-normal distribution with a tiny negative skew. Ultimately, the scatter plot of lagged versus real return, which is colored by directional class labels, demonstrates significant class overlap and no obvious structure, supporting the idea that lagged return is not a discriminative tool on its own.

Graph 1: Time Series Plot



Histogram 1: IVV Daily Returns





4. Conclusion

This outcome shows that single, raw technical indicators such as lagged return do not reliably predict price direction in developed markets and further implies that a good selection of features can improve the efficiency of the computational resources and attain better prediction results. The authors used LASSO (Least Absolute Shrinkage and Selection Operator) to choose features from a pool of more complex and parameterized technical indicators, as was highlighted in the original work. A crucial step in improving model performance was emphasized as being indicator tuning, which involves modifying parameters like window size or smoothing constants (Sagaceta Mejia et al.).

This result has a number of financial ramifications. First, it is consistent with earlier research that cautions against relying too much on basic technological criteria. Second, by showing that feature optimization and model sophistication are prerequisites for deriving significant signals from market data, it confirms the methods used in the Sagaceta Mejia et al. study. Last but not least, the results also point to possible distinctions between developed and emerging markets: developed markets, such as the U.S., are far more efficient, and any patterns that could be exploited are likely to be transient or masked by noise, whereas inefficiencies in emerging markets may allow some indicators to maintain predictive power.

To sum up, our replication validates the significance of the multi-step modeling process employed by the original authors and acts as a thorough evaluation of a minimum model. It emphasizes how technical indicators by themselves only produce weak signals in the absence of

proper feature engineering, parameter adjustment, and sophisticated modeling, particularly in efficient markets like the S&P 500.

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